BOUNDEDNESS AND COHERENCE PROPERTIES PRESERVED BY THE UPPER SHRIEK OF A FINITE TYPE HOMOMORPHISM

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The preservation of bounded below coherent complexes by the derived internal hom along a finite ring map. Invoked inside the counted proof (source lines 4646-4648). Supplies the bounded below and coherent case for the internal hom factor of the shriek.
- The description of the restriction of the internal hom as the tensor with a perfect complex in the perfect case. Invoked inside the counted proof (source lines 4652-4654). Identifies the internal hom with the tensor product with a perfect complex whose representatives are finite complexes of finite projectives.
- The perfectness of a module over a polynomial ring in the perfect case of a finite type homomorphism. Invoked inside the counted proof (source lines 4649-4650). Provides the hypothesis that makes the tensor description available.

Source reference. Original lemma lemma-exact-support-coherent: sources/stacks-dualizing/pinned-original-source.tex, source lines 2480–2480. Navigation pointer only: the counted text refers to the Stacks item lemma-exact-support-coherent, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original reference more-algebra-lemma-dual-perfect-complex: sources/stacks-dualizing/pinned-original-source.tex, source lines 4653–4653. Navigation pointer only: the counted text refers to the Stacks item more-algebra-lemma-dual-perfect-complex, which belongs to another chapter of the project and is not part of this

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Source reference. Original reference more-algebra-lemma-perfect-ring-map: sources/stacks-dualizing/pinned-original-source.tex, source lines 4650–4650. Navigation pointer only: the counted text refers to the Stacks item more-algebra-lemma-perfect-ring-map, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $\varphi : R \to A$ be a finite type homomorphism of Noetherian rings.*

(1) *$\varphi^!$ maps $D^+(R)$ into $D^+(A)$ and $D^+_{Coh}(R)$ into $D^+_{Coh}(A)$.*
(2) *if φ is perfect, then $\varphi^!$ maps $D^-(R)$ into $D^-(A)$, $D^-_{Coh}(R)$ into $D^-_{Coh}(A)$, and $D^b_{Coh}(R)$ into $D^b_{Coh}(A)$.*

Proof. Choose a factorization $R \to P \to A$ as in the definition of $\varphi^!$. The functor $-\otimes^{\mathbf{L}}_R : D(R) \to D(P)$ preserves the subcategories $D^+, D^+_{Coh}, D^-, D^-_{Coh}, D^b_{Coh}$. The functor $R\mathrm{Hom}(A, -) : D(P) \to D(A)$ preserves D^+ and D^+_{Coh} by Lemma lemma-exact-support-coherent. If $R \to A$ is perfect, then A is perfect as a P-module, see More on Algebra, Lemma more-algebra-lemma-perfect-ring-map. Recall that the restriction of $R\mathrm{Hom}(A, K)$ to $D(P)$ is $R\mathrm{Hom}_P(A, K)$. By More on Algebra, Lemma more-algebra-lemma-dual-perfect-complex we have $R\mathrm{Hom}_P(A, K) = E \otimes^{\mathbf{L}}_P K$ for some perfect $E \in D(P)$. Since we can represent E by a finite complex of finite projective P-modules it is clear that $R\mathrm{Hom}_P(A, K)$ is in $D^-(P), D^-_{Coh}(P), D^b_{Coh}(P)$ as soon as K is. Since the restriction functor $D(A) \to D(P)$ reflects these subcategories, the proof is complete. □